Roll No. : ....

MID SEMESTER EXAMINATION SEPT. 2025

Name of the Program: **B.Tech** Semester: V

Name of the Course : Bigdata Visualization Course Code: TCS571

Time : 90 Minutes **Maximum Marks : 50**

Note:

i. Answer all the questions by choosing any one of the sub questions
ii. Each questions carries 10 marks

Q1	(10 Marks)	
(a)	Elaborate the Eight Hats approach for designing data visualizations. How does this methodology help in mitigating visualization errors? Support your explanation with practical examples.	CO1
	OR	
(b)	Outline the step-by-step process for developing a successful data visualization project. Illustrate each stage with relevant examples.	CO1
Q2	(10 Marks)	
(a)	Identify and discuss the primary challenges related to achieving accurate interpretation in data visualizations. What strategies can be employed to overcome these issues? Explain in detail.	CO2
	OR	
(b)	Provide a critical assessment of the visualization anatomy framework. In what ways does this framework contribute to producing effective data visualizations?	CO2
Q3	(10 Marks)	
(a)	Explain how a stacked bar chart can effectively represent part-to-whole relationships within a dataset. Present a real-world example to illustrate your answer.	CO2
	OR	
(b)	Distinguish between data transformation and data cleaning, emphasizing the significance of each in the context of visualization design.	CO2
Q4	(10 Marks)	
(a)	Describe the essential steps involved in preparing and refining data for visualization. Explain the impact of physical data properties on the effectiveness of a visualization.	CO3
	OR	
(b)	Analyze the connection between visualization accuracy and the level of trust the audience places in the data presented. Use examples to support your discussion.	CO3
Q5	(10 Marks)	
(a)	Discuss the concept and application of a pixelated bar chart, including an example. Evaluate why selecting the proper chart size is crucial for accurately representing data.	CO2
	OR	
(b)	Given a complex dataset of weather forecasts, describe in detail how you would prepare and refine the data, select appropriate visualization techniques, integrate design metaphors, and ultimately communicate the underlying story effectively.	CO3